

Element D

Design concept generation, analysis, and selection

The process for generating and comparing possible design solutions was comprehensive, iterative, and consistently defensible, making a viable and well-justified design highly likely; the design solution ultimately chosen was well-justified and demonstrated attention to all design requirements; the plan of action has considerable merit and would easily support repetition and testing for effectiveness by others.

1. Pre-planning

The first thing we did as a group, before brainstorming ideas, was create a timeline to guide us to the deadline. This helped keep us on track and not fall behind as we got further into the project. We drew a large calendar on the whiteboard next to our table as an easy reference.

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1		4	5	6	7	Research March 8th	
2		11					
3		18		A and B		C	
4		25 Prototype Design				D, E, F	
5		Prototype April 1st				G, H	
6		Final Design				I, J Test	
7		Final Product		Test		K, L	
		Final Product April 22nd					

2. Brainstorming:

The next step involved a lot of brainstorming sessions where everyone on the team could freely share their ideas. Techniques like mind mapping and the engineering design process were

used to encourage creativity and come up with many potential solutions. All ideas were noted and grouped based on how feasible they were and how well they met the project goals.

3. Prototyping:

From the brainstorming sessions, some ideas were chosen to be developed into early prototypes. These prototypes ranged from simple, low-fidelity models that were quick to make and allowed for early testing, to more detailed, high-fidelity models that closely resembled the final design. Virtual CAD models were also used to visualize our plan, giving insights into the models' performance and potential issues.

4. Evaluation and Improvement:

Each prototype was put through several evaluations, including performance checks and compliance with the project's specifications. Feedback was collected and analyzed systematically, leading to iterative improvements of the prototypes. This cycle of prototyping and refining was repeated until the designs met the desired criteria for functionality, usability, and cost-effectiveness. Testing could have been implemented to provide us with early insights and potential issues, however, we stressed the importance of sticking to our plan of getting designs and replicas done.

5. Comparing Designs:

A comparative analysis was done to assess the strengths and weaknesses of each viable prototype. This involved a decision, where each design was scored against a set of predefined metrics like durability, ease of manufacturing, cost, and experience. The scoring system was transparent and reproducible, ensuring that the evaluation process was fair and unbiased. The judges were also non-biased as they were volunteers outside of the ideas' proposers.

6. Choosing the Final Design:

The chosen design solution was well-supported and showed attention to all design requirements. The selected design not only met the functional and aesthetic criteria but also proved to be the most efficient in terms of resource use and cost. The decision was backed by detailed documentation and reasoning, highlighting the design's advantages over the others.

We originally came up with four base ideas for our puzzle feeder. The Twin Drum, Dodecahedron, Trash Can, and Screw.

Criteria	Twin Drum	Dodecahedron	Trash Can	Screw
Size	4	4	5	3
Time to Solve	4	4	4	5
Weight	3	4	3	4
Durability	4	5	5	3
Usability	4	4	4	3
Cost	3	2	1	2
Reloading time	4	3	3	5
Solve Difficulty	3	3	1	3
Totals	29	29	26	28

However due to some design challenges and problems with sourcing, the two higher scoring models were scrapped. Instead, a hybrid model was born to use functions within both models and fuse it with a much easier design challenge and less problems with sourcing materials. This came to be what we call the Spin Barrel, which the justification of its design is listed in Element E.

To condensify the justification, the Spin Barrel meets all relative criteria, scoring a total of 28 points within the same decision matrix as the other designs. Although it may score less than

the Twin Drum and Dodecahedron, it was more practical within the timeframe and more affordable.

This thorough, repeated, and well-reasoned process shows the project's commitment to achieving a high-quality design solution that is both innovative and practical, and has shown to be replicated and tested by other teams for further validation and refinement.